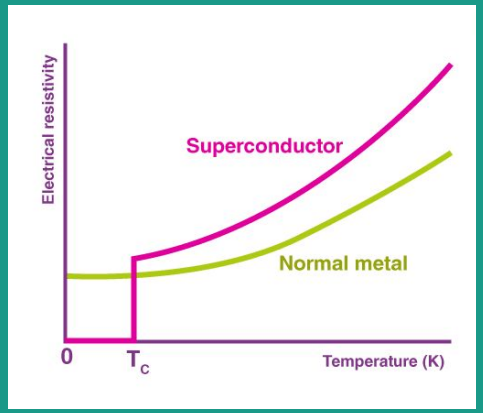
Industrial Superconducting Quantum Computers

Michael Silver - UTQC qLearn Week 4

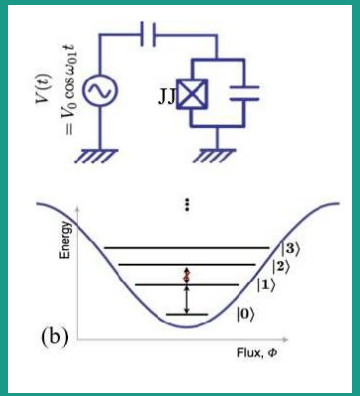


Last Week

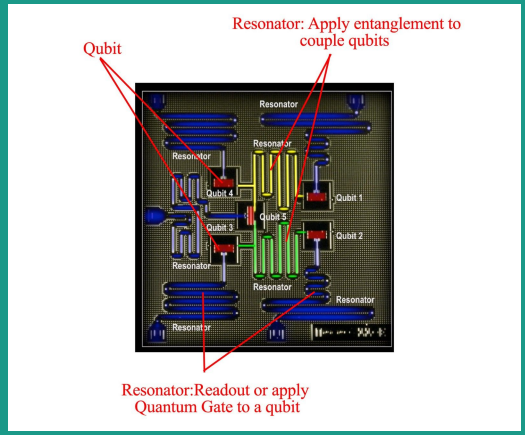
What is a Superconductor?



Superconducting Qubits



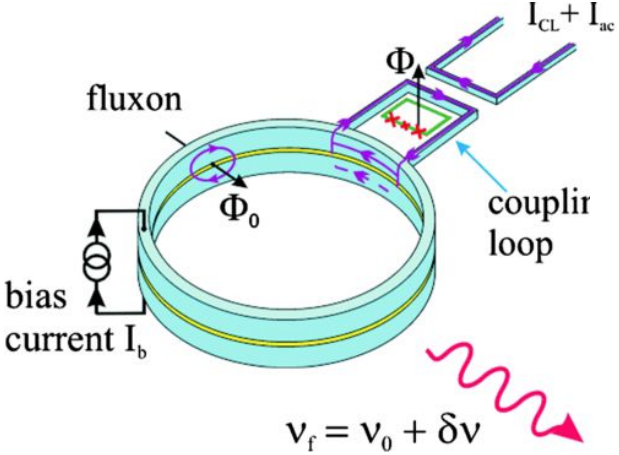
Superconducting Computer Design



Types of Superconducting Qubits

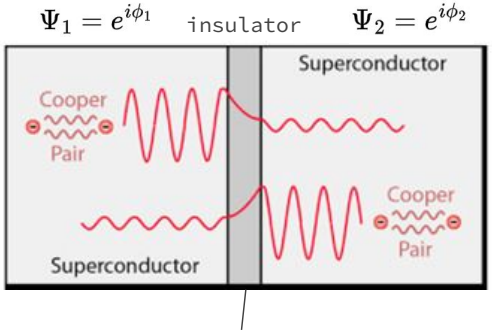
Flux Qubit

Information encoded in direction of current



Phase Qubit

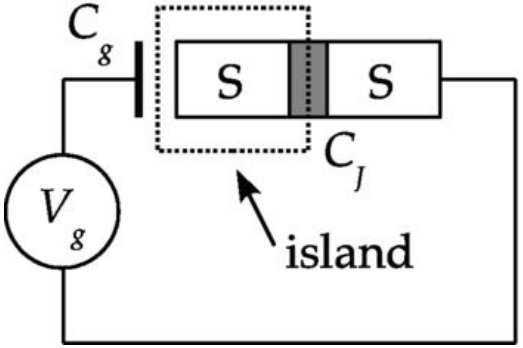
Encodes info in the phase difference across the JJ



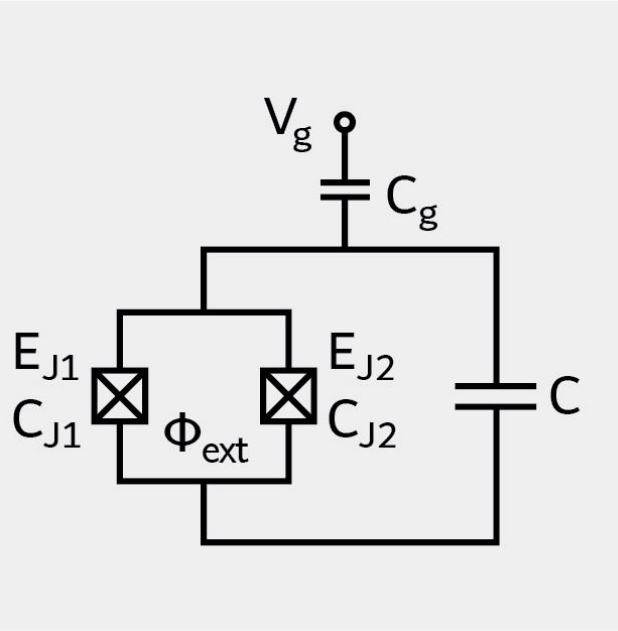
Quantum Tunneling Effect

Charge Qubit (Transmon)

Encodes info in the number of Cooper pairs on the island



Tunable Transmons using SQUIDs



- **SQUID** (Superconducting Quantum Interference Device): Superconducting loop with two Josephson junctions; modifies current of loop based on external magnetic field
- **Why do we care?**
  - inductance of JJ prop to loop current $L_j \propto \frac{1}{I_c}$
  - Qubit resonant frequency prop to JJ inductance

$$\omega_{01} \approx \frac{1}{\sqrt{L_J C}}$$

Therefore tunable qubits! Separating qubits by frequency, then bringing them together for two-qubit gates, makes a system less susceptible to crosstalk!

‘Physical’ Qubits

T_1: Relaxation time (Moving from 1 to 0)

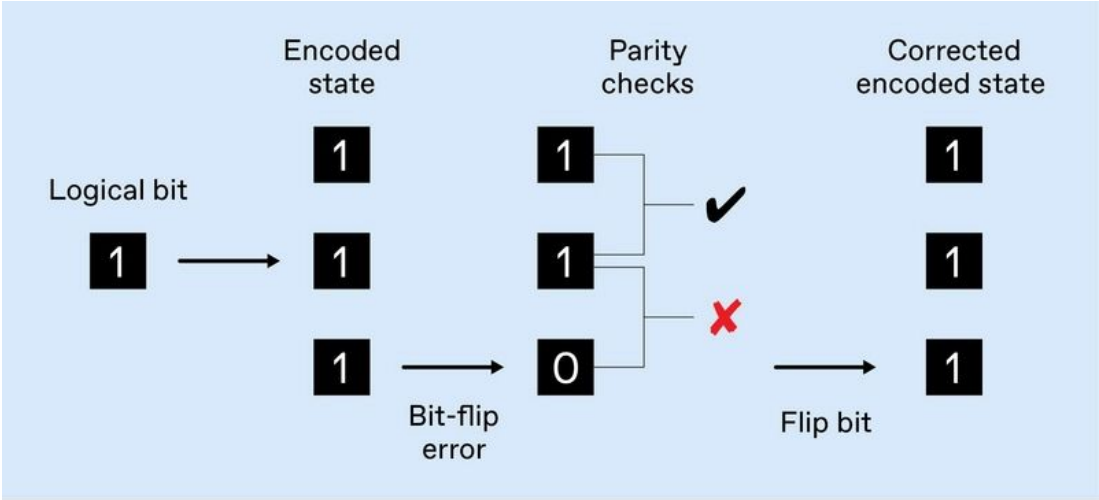
T_2: Dephasing time (loses phase information)

Why?? ENTROPY

quantum.cloud.ibm.com

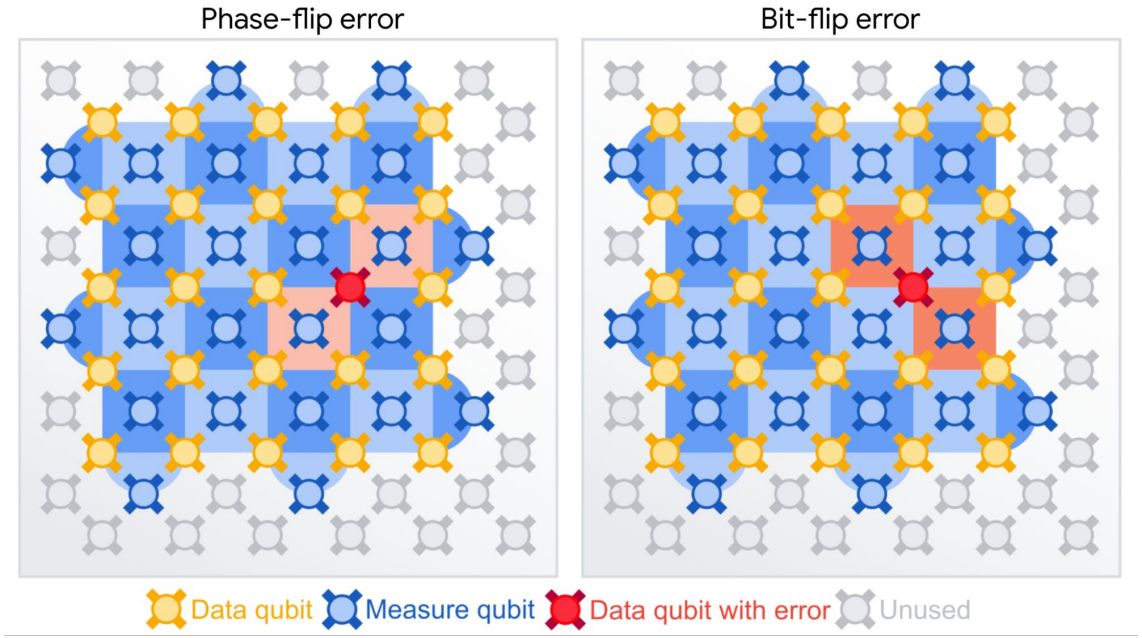
Fault-Tolerant QC

- Protecting against all errors physically is nearly impossible (Mario); we use ERROR CORRECTION
- Classical Error Correction scheme below: why wouldn't it work for quantum?



Quantum Error Correct: Surface Codes!

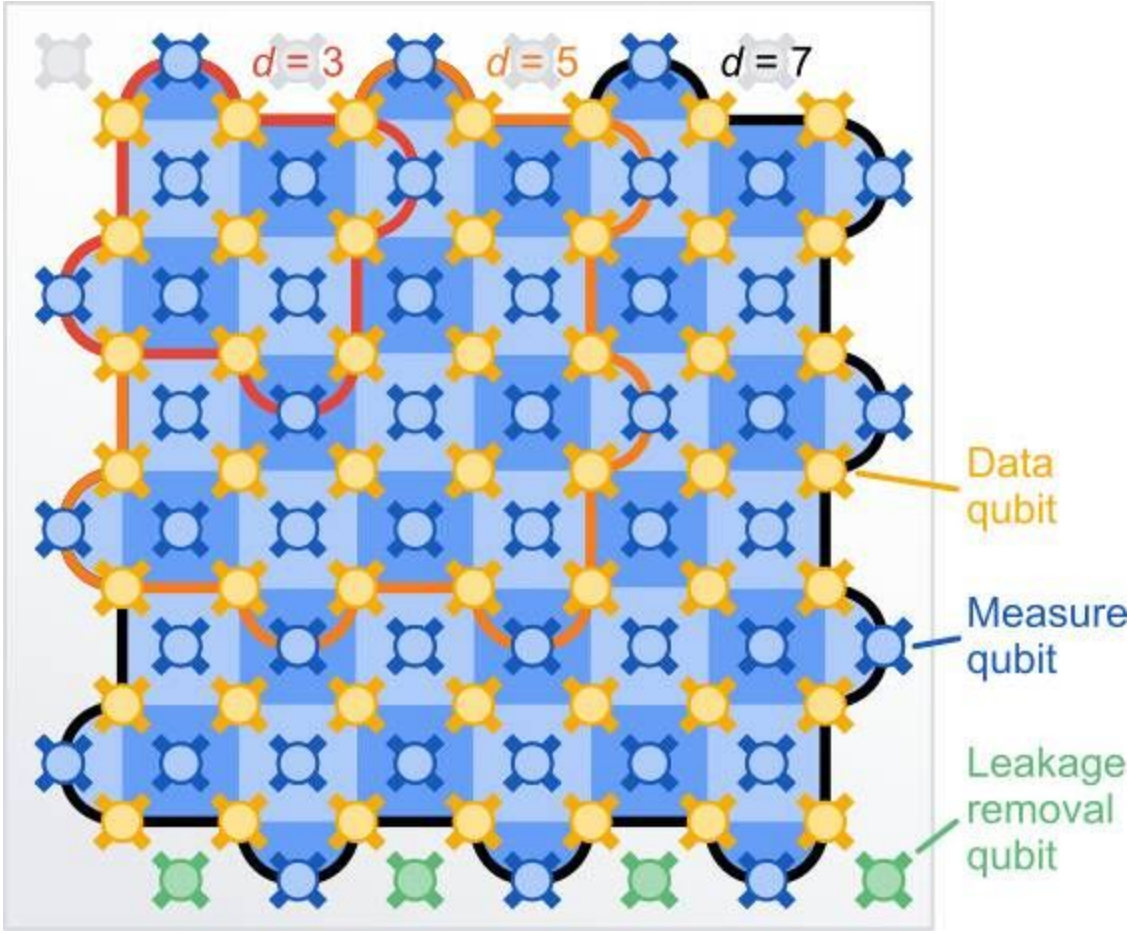
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \rightarrow \alpha|000\rangle + \beta|111\rangle$$



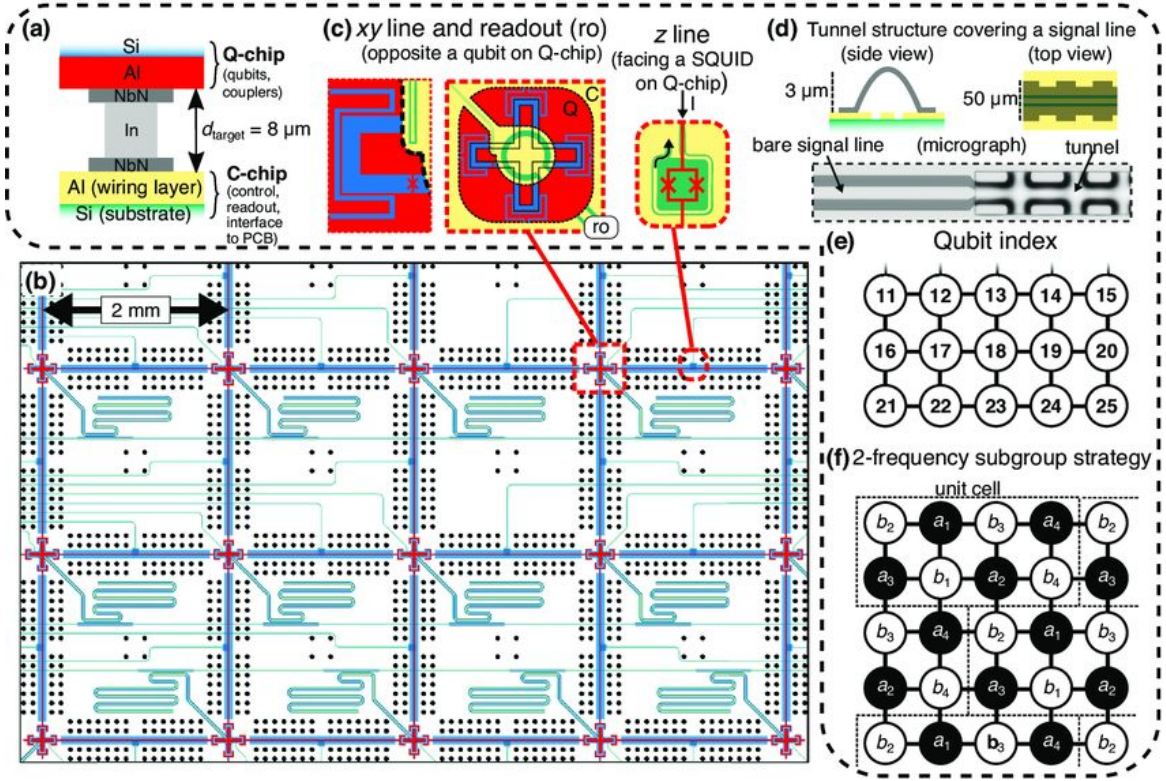
Quantum Error Correct: Surface Codes!

- **Logical Qubits:** Error-corrected qubits that act as our things we do operations on
- **Parity Operator:** Operation we use to check if qubits are in matching or opposing states
  - **Stabilizers:** Specific parity operator chosen to define/protect logical qubit state
- **Syndromes:** The measurements we make to our *ancillas* to diagnose errors
  - If you're reading this like a good student should, what is an ancilla?

Curious about what these are? Attend our algorithms lectures!

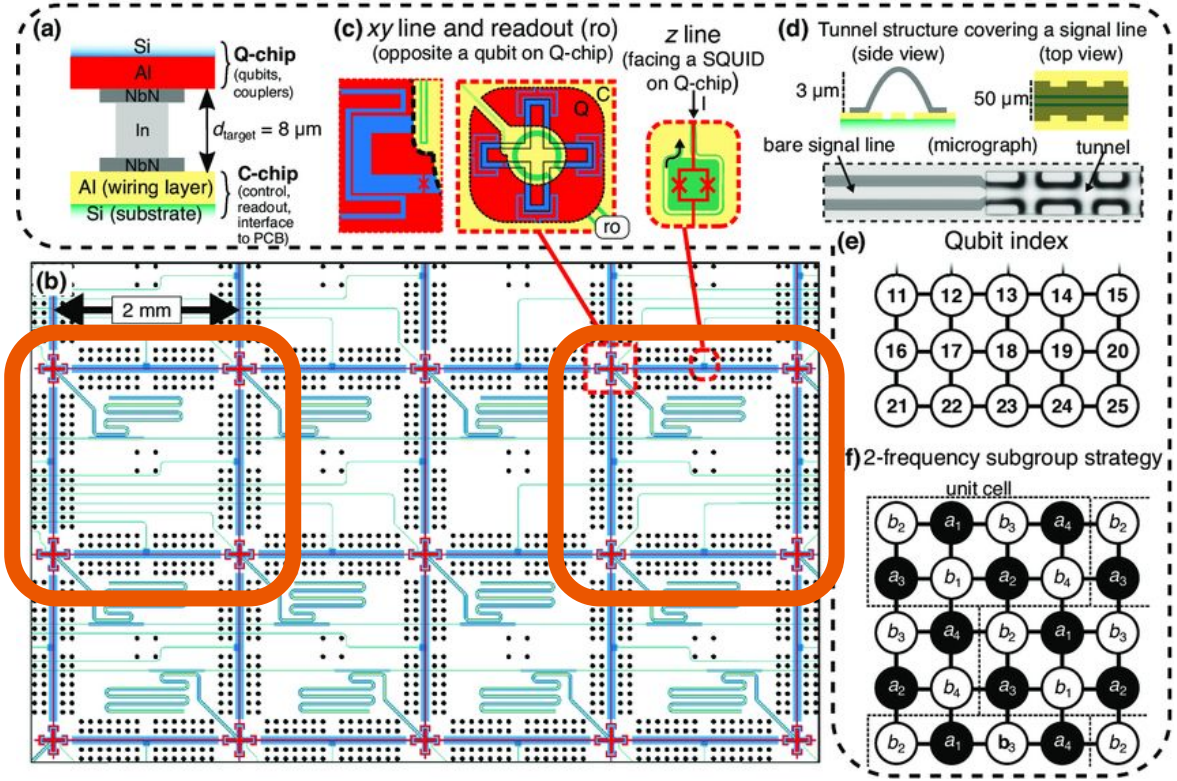

d = 3
d = 5
d = 7
Data qubit
Measure qubit
Leakage removal qubit

Chip Connectivity (nearest neighbour!)



Chip Connectivity (nearest neighbour!)

How would these two logical qubits talk to each other if they don't share a microwave resonator link?



Lattice Surgery

- ‘In-between’ transmons (syndrome transmons) are off initially, separating all our logical qubits
- When we want logical qubits to talk to each other, we ‘merge’ them by turning on our syndrome transmons and start measuring parity operators across both qubits
- By measuring parity across the seam, the two separate grids essentially ‘stitch’ together
- Once our operation is finished, we turn off the syndrome transmons again

Lattice Surgery

